A Feng–Tian–Wei-type central limit theorem for the high-temperature SYK partition function

Abstract

Fix an even interaction order $q \geq 4$ and let $Z_N(\beta)$ be the normalized-trace partition function of the Gaussian SYK model. We prove, for all sufficiently small fixed real $\beta \neq 0$,

$$\frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\sqrt{\operatorname{Var} Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1).$$

The proof is a high-temperature modification of the argument of Feng–Tian–Wei. Their fixed-test-function proof isolates the radial statistic $M_N^{-1}\sum_I (g_I^2 - 1)$ by subtracting a multiple of x^2. For the N-dependent test function representing the partition function, we replace that asymptotic subtraction by an exact second-Wiener-chaos projection. The remaining chaoses are controlled in two ranges: an FTW-type parity count up to order $A \log N$, and analytic suppression above $A \log N$. The proof uses two external input packages: recent complex second-moment and zero-free estimates, and the high-temperature annealed free-energy limit together with its Gaussian–Rademacher transfer. Every further deduction from these packages is included below.

1 Model and main results

1.1 The SYK model

Fix an even integer $q \geq 4$, and let N be even. Let

$$\mathcal{I}_N = \{I \subset [N] : |I| = q\}, \qquad M_N = \binom{N}{q}, \qquad \sigma_N^2 = \frac{(q-1)!}{N^{q-1}},$$

and let $(g_I)_{I \in \mathcal{I}_N}$ be independent standard Gaussian random variables. Let $\psi_1, \ldots, \psi_N$ be Majorana matrices satisfying $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$. For $I = \{i_1 < \cdots < i_q\}$ set

$$\psi_I = i^{q/2}\psi_{i_1} \cdots \psi_{i_q}.$$

Thus ψ_I is Hermitian, $\psi_I^2 = \mathbb{I}$, and $\|\psi_I\| = 1$. Put

$$H_N(g) = \sigma_N \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad Z_N(z; g) = \operatorname{tr} e^{-zH_N(g)}, \qquad F_N(z) = \mathbb{E}Z_N(z; g).$$

Here $\operatorname{tr} = 2^{-N/2}\operatorname{Tr}$ is the normalized trace, so $Z_N(0) = F_N(0) = 1$. Throughout, all limits are taken as $N \to \infty$ through even integers.

1.2 Main results

Theorem 1.1 (Partition-function CLT)**.** *Fix a real β satisfying*

$$0 < |\beta| < \min\{\beta_{\mathrm{FE}}, R_q\}, \qquad \beta \ell_q'(\beta) \neq 0.$$

Then

$$N^{q/2-1}\left(\frac{Z_N(\beta)}{\mathbb{E}Z_N(\beta)} - 1\right) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi, \qquad \xi \sim \mathcal{N}(0,1). \tag{1}$$

Moreover,

$$N^{q-2}\frac{\operatorname{Var} Z_N(\beta)}{(\mathbb{E}Z_N(\beta))^2} \longrightarrow \frac{q!}{2}\,\beta^2\ell_q'(\beta)^2, \tag{2}$$

and hence

$$\frac{Z_N(\beta)-\mathbb{E}Z_N(\beta)}{\sqrt{\operatorname{Var} Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1). \tag{3}$$

Corollary 1.2 (Free-energy CLT)**.** *Under the assumptions of Theorem 1.1,*

$$N^{q/2-1}\left(\log Z_N(\beta)-\mathbb{E}\log Z_N(\beta)\right) \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi, \qquad \xi\sim\mathcal{N}(0,1).$$

Proof. Put

$$a=\frac{q}{2}-1, \qquad X_N=\frac{Z_N(\beta)-\mathbb{E}Z_N(\beta)}{\mathbb{E}Z_N(\beta)}, \qquad W_N=\log\frac{Z_N(\beta)}{\mathbb{E}Z_N(\beta)}.$$

Thus $W_N=\log(1+X_N)$, and Theorem 1.1 gives

$$N^aX_N \Longrightarrow \sqrt{\frac{q!}{2}}\,\beta\ell_q'(\beta)\,\xi, \qquad \delta_N^2:=\mathbb{E}X_N^2=O(N^{-2a}).$$

We first record the estimate needed to replace the deterministic centering. The function $g\mapsto W_N(g)$ is Lipschitz, uniformly in N. Indeed,

$$\partial_{g_I}W_N=-\beta\sigma_N\langle\psi_I\rangle_{\beta,g}.$$

For each $(q-1)$-set S, the operators $(\psi_{S\cup\{i\}})_{i\notin S}$ are pairwise anticommuting Hermitian involutions. The usual anticommuting-observable estimate, summed over S, therefore gives

$$\sum_{I\in\mathcal{I}_N}|\langle\psi_I\rangle_{\beta,g}|^2\le\frac{1}{q}\binom{N}{q-1}.$$

Consequently,

$$\|\nabla_gW_N\|_2^2\le\frac{\beta^2\sigma_N^2}{q}\binom{N}{q-1}\le\frac{\beta^2}{q}.$$

Gaussian concentration now shows that $W_N-\mathbb{E}W_N$ is sub-Gaussian with a constant depending only on q and β.

For completeness, this concentration bound and $\mathbb{E}e^{W_N}=1$ imply

$$-\frac{\beta^2}{2q}\le\mathbb{E}W_N\le 0.$$

Moreover, with $h(w)=e^w-1-w$, one has $h(w)\le C(e^w-1)^2$ for $w\ge-1$ and $h(w)\le|w|$ for $w<-1$. Chebyshev's inequality gives

$$\mathbb{P}(W_N<-t)\le\frac{\delta_N^2}{(1-e^{-t})^2}\le C\delta_N^2, \qquad t\ge 1,$$

whereas Gaussian concentration gives a Gaussian upper bound for the same probability when t is large. Integrating the minimum of these two bounds yields

$$0\le-\mathbb{E}W_N=\mathbb{E}h(W_N)\le C\delta_N^2\sqrt{\log(1/\delta_N)}=O\Big(N^{2-q}\sqrt{\log N}\Big)=o(N^{-a}).$$

Finally, N^aX_N is tight and $X_N\to0$ in probability. Hence the Taylor estimate $|\log(1+x)-x|\le Cx^2$ for $|x|\le1/2$ gives

$$N^a\big(W_N-X_N\big)\longrightarrow 0 \qquad \text{in probability.}$$

Since

$$\log Z_N-\mathbb{E}\log Z_N=W_N-\mathbb{E}W_N,$$

the claimed convergence follows from Theorem 1.1 and Slutsky's theorem. □

1.3 Outline of the proof

It is useful to record the normalization used by Feng–Tian–Wei. Define

$$\widehat{H}_N(g) = M_N^{-1/2} \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad \tau_N = \sigma_N \sqrt{M_N}. \tag{4}$$

Then $H_N = \tau_N \widehat{H}_N$ and

$$\tau_N^2 = \sigma_N^2 M_N = \frac{N}{q}\big(1 + O_q(N^{-1})\big). \tag{5}$$

Consequently, if $\widehat{L}_N(f) = \operatorname{tr} f(\widehat{H}_N)$, then

$$Z_N(\beta) = \widehat{L}_N(f_{N,\beta}), \qquad f_{N,\beta}(x) = e^{-\beta \tau_N x}. \tag{6}$$

Set $a = q/2 - 1$ and normalize the partition function by $Y_N = Z_N / F_N$. Clifford sign-gauge symmetry eliminates the first Gaussian chaos and identifies the second chaos exactly:

$$Y_N(\beta) - 1 = A_N(\beta) Q_N + \mathcal{R}_N(\beta), \qquad A_N(\beta) = \frac{\beta}{2} \partial_\beta \log F_N(\beta), \qquad Q_N = \frac{1}{M_N} \sum_I (g_I^2 - 1).$$

Thus the leading random term is a normalized sum of independent centered χ_1^2 variables. The main task is to prove

$$\|\mathcal{R}_N(\beta)\|_2^2 = o(N^{-2a}).$$

For this purpose, split the chaos expansion at $m_N = \lceil A \log N \rceil$. Up to order m_N, a Duhamel derivative bound and the sign-gauge parity rule reduce the estimate to counting even tuples of q-sets. Such a tuple of length r uses at most $qr/2$ vertices, which makes the sum of chaoses of orders $3, \dots, m_N$ equal to $o(N^{-2a})$. Above m_N, the r-th chaos vanishes to order at least r at $\beta = 0$. Applying an L^2-valued analytic estimate on a zero-free disk, together with the complex second-moment input, gives a geometric factor $(|\beta|/R)^{2m_N}$. Choosing A sufficiently large yields the same $o(N^{-2a})$ bound for the remaining chaoses.

The annealed free-energy input and convexity imply

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2} \ell_q'(\beta).$$

Since $M_N \sim N^q/q!$, the ordinary central limit theorem for Q_N, the remainder estimate, and Slutsky's theorem give the claimed limit for $N^a(Y_N - 1)$. Orthogonality of distinct Gaussian chaoses gives the matching variance asymptotic and hence the standardized central limit theorem.

2 Preliminaries

2.1 A complex second-moment estimate

We use the following recent high-temperature preprint result. It is Theorems 3 and 4 of [2], with the deterministic change of inverse-temperature coming from the Majorana normalization. In particular, the constants below depend only on q.

Proposition 2.1 (Complex second-moment input)**.** *There are constants $R_q > 0$, $C_q < \infty$, and $N_q < \infty$ such that, for every even $N \geq N_q$,*

$$F_N(z) \neq 0 \qquad (|z| \leq R_q), \tag{7}$$

and

$$\sup_{|z| \leq R_q} \mathbb{E} \left| \frac{Z_N(z;g)}{F_N(z)} - 1 \right|^2 \leq C_q N^{1-q/2}. \tag{8}$$

Proof. This is Theorems 3 and 4 of [2], specialized to the unperturbed model, followed only by a deterministic normalization change. That paper uses Majoranas satisfying $\{\psi_i^{\mathrm{Zlo}}, \psi_j^{\mathrm{Zlo}}\} = \delta_{ij}\mathbb{I}$. Thus $\psi_i^{\mathrm{Zlo}} = 2^{-1/2}\psi_i$, $H_N^{\mathrm{Zlo}} = 2^{-q/2}H_N$, and

$$Z_N^{\mathrm{ours}}(z) = Z_N^{\mathrm{Zlo}}(2^{q/2}z).$$

Normalized versus unnormalized trace cancels in the ratio in (8). We choose R_q so that $\{|2^{q/2}z| \le R_q^{\mathrm{Zlo}}\}$ lies inside the zero-free and second-moment region in those theorems. This gives (7)–(8) for all sufficiently large N. □

Set

$$a = \frac{q}{2} - 1 > 0, \qquad Y_N(z;g) = \frac{Z_N(z;g)}{F_N(z)}.$$

Thus $\mathbb{E}Y_N(z) = 1$, and the right-hand side of (8) is $C_q N^{-a}$. Let P_r denote the orthogonal projection in complex $L^2(\gamma_N)$ onto the r-th homogeneous Gaussian chaos, where γ_N denotes standard Gaussian measure on $\mathbb{R}^{M_N}$. We write $\|X\|_2^2 = \mathbb{E}|X|^2$. We use the probabilists' Hermite polynomials

$$H_\alpha(g) = \prod_{I \in \mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I), \qquad \mathbb{E}[H_\alpha H_\eta] = \alpha!\, \mathbf{1}_{\{\alpha=\eta\}},$$

where $\alpha! = \prod_I \alpha_I!$. For every sufficiently regular random variable X,

$$\|P_r X\|_2^2 = \sum_{|\alpha|=r} \frac{|\mathbb{E}\, \partial_g^\alpha X|^2}{\alpha!}. \tag{9}$$

Lemma 2.2 (L^2-valued analyticity)**.** *For each fixed N, the map $z \mapsto Z_N(z;\cdot)$ is entire with values in $L^2(\gamma_N)$. Hence $z \mapsto Y_N(z;\cdot)$ is $L^2(\gamma_N)$-valued analytic on every zero-free domain of F_N.*

Proof. Since $\|\psi_I\| = 1$,

$$\|H_N(g)\| \le \sigma_N \sum_{I \in \mathcal{I}_N} |g_I|.$$

For $|z| \le R$ and every $k \ge 0$, the spectral theorem gives

$$|\partial_z^k Z_N(z;g)| \le \|H_N(g)\|^k\, e^{R\|H_N(g)\|}.$$

The square of the right-hand side is integrable because the Gaussian vector is finite dimensional. Dominated convergence therefore permits differentiation in L^2 on every compact disk. Division by the nonvanishing deterministic analytic function F_N proves the second assertion. □

2.2 Deterministic FTW coefficient

The remainder has now been controlled. To identify the deterministic coefficient $A_N(\beta)$ in (16), we use the recent preprint result on the annealed free-energy limit. The derivative convergence needed below is not taken as an additional input; it follows from convexity.

Lemma 2.3 (Convex convergence and derivatives)**.** *Let $I \subset \mathbb{R}$ be open. Suppose that $h_N : I \to \mathbb{R}$ are finite, differentiable, convex functions and $h_N(x) \to h(x)$ pointwise, where $h \in C^1(I)$. Then $h_N \to h$ and $h_N' \to h'$ locally uniformly on I.*

Proof. Fix $[a,b] \Subset I$ and choose $u < a < b < v$ in I. Monotonicity of secant slopes bounds every secant slope of h_N on $[a,b]$ between quantities formed from $h_N(u), h_N(a), h_N(b), h_N(v)$. These endpoint values converge, so the h_N are equi-Lipschitz on $[a,b]$. Pointwise convergence on a finite mesh, followed by mesh refinement, gives uniform convergence there.

For $x \in I$ and sufficiently small $s > 0$, convexity gives

$$\frac{h_N(x) - h_N(x-s)}{s} \le h_N'(x) \le \frac{h_N(x+s) - h_N(x)}{s}.$$

First let $N \to \infty$ and then $s \downarrow 0$ to obtain $h_N'(x) \to h'(x)$. Finally, h_N' are nondecreasing. On a fine partition of $[a,b]$, monotonicity traps $h_N'(x)$ between its values at adjacent mesh points, while uniform continuity traps $h'(x)$ between the corresponding values of h'. Pointwise convergence at the finitely many mesh points then gives uniform convergence of the derivatives. □

Proposition 2.4 (High-temperature annealed free energy)**.** *There are $\beta_{\mathrm{FE}} = \beta_{\mathrm{FE}}(q) > 0$ and an even real-analytic function ℓ_q on $(-\beta_{\mathrm{FE}}, \beta_{\mathrm{FE}})$ such that, locally uniformly on this interval,*

$$\frac{1}{N} \log F_N(\beta) \longrightarrow \ell_q(\beta), \tag{10}$$

$$\frac{1}{N} \partial_\beta \log F_N(\beta) \longrightarrow \ell_q'(\beta). \tag{11}$$

Moreover,

$$\ell_q(\beta) = \frac{\beta^2}{2q} + O(\beta^4), \qquad \ell_q'(\beta) = \frac{\beta}{q} + O(\beta^3). \tag{12}$$

Proof. The pointwise convergence in (10) is Theorem 5 of [3], together with the Gaussian extension stated there. That extension invokes the Gaussian–Rademacher universality result [4]; we use it as part of the free-energy input package.

We now separate the normalization conversion from this transfer. The Majoranas of [3] satisfy

$$\psi_i^{\mathrm{GPSZ}} = 2^{-1/2} \psi_i, \qquad H_N^{\mathrm{GPSZ}} = 2^{-q/2} H_N.$$

Set $u = 2^{q/2}\beta$. Gaussian symmetry gives

$$\mathbb{E} \operatorname{tr} e^{-\beta H_N} = \mathbb{E} \operatorname{tr} e^{+\beta H_N},$$

and $u H_N^{\mathrm{GPSZ}} = \beta H_N$. Thus the sign and normalization conventions of [3] give

$$\frac{1}{N} \log F_N(\beta) = \frac{u}{N} f_{N,q}^{\mathrm{GPSZ}}(u), \tag{13}$$

where $f_{N,q}^{\mathrm{GPSZ}}(u) = u^{-1} \log \mathbb{E} Z_N^{\mathrm{GPSZ}}(u)$.

For completeness, we also extract the regularity and the first coefficient from the explicit formula in that theorem. In its notation,

$$\Phi_u(z) = \sum_{v \ge 1} \mu_v(u) z^v, \qquad u f_q^{\mathrm{GPSZ}}(u) = \Phi_u(t_*(u)) - 1 - \log t_*(u),$$

where $t_* \partial_z \Phi_u(t_*) = 1$. A contributing hypertree has $v = 1 + (q-1)s$ vertices and its coefficient contains

$$\rho(u)^{v-1}, \qquad \rho(u)^{q-1} = \frac{(q-1)! u^2}{2^{q+1}}.$$

More explicitly, $\mathcal{T}_v^{\mathrm{sim}}$ is empty unless $v = 1 + (q-1)s$ for some $s \ge 0$. For $s \ge 1$ we may therefore write

$$\mu_{1+(q-1)s}(u) = c_{q,s} u^{2s},$$

where

$$c_{q,s} = \frac{1}{(1+(q-1)s)!} \left(\frac{(q-1)!}{2^{q+1}} \right)^s \sum_{T \in \mathcal{T}_{1+(q-1)s}^{\mathrm{sim}}} b_T.$$

The bounds used in the proof of Proposition 3 of [3] give $|c_{q,s}| \le C_q^s$. Hence

$$\Phi_u(z) = z + \sum_{s\ge 1} c_{q,s} u^{2s} z^{1+(q-1)s}$$

converges normally for (u^2, z) in a neighborhood of $(0,1)$. Thus $\Phi_u(z)$ is jointly analytic in (u^2, z) there. At $u = 0$, $\Phi_0(z) = z$ and $t_*(0) = 1$; the derivative with respect to t of $t\partial_z \Phi_u(t) - 1$ is 1 at $(u,t) = (0,1)$. The analytic implicit function theorem makes t_*, and hence the limiting right-hand side, analytic in u^2.

The first nontrivial term is the unique one-edge hypertree on q vertices:

$$\Phi_u(z) = z + \frac{u^2}{q2^{q+1}} z^q + O(u^4).$$

It follows successively that

$$t_*(u) = 1 - \frac{u^2}{2^{q+1}} + O(u^4), \qquad u f_q^{\mathrm{GPSZ}}(u) = \frac{u^2}{q2^{q+1}} + O(u^4).$$

Using $u = 2^{q/2}\beta$ in (13) gives (12).

It remains only to justify the two local-uniform assertions. Let $h_N = N^{-1} \log F_N$. If

$$\overline{\nu}_N(B) = \mathbb{E}\ \mathrm{tr}\, \mathbf{1}_{\{H_N \in B\}},$$

then $F_N(\beta) = \int e^{-\beta x} \overline{\nu}_N(\mathrm{d}x)$; hence h_N is finite, differentiable, and convex. Differentiation is justified by the Gaussian exponential-integrability bound in Lemma 2.2. The limiting function just obtained is C^1. Lemma 2.3 now proves both local-uniform convergences. □

By (16) and (11),

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2} \ell_q'(\beta). \tag{14}$$

3 Proof of the main result

For a fixed test function f, Feng–Tian–Wei [1] prove the Gaussian SYK linear-statistics CLT by isolating

$$Q_N := \mathrm{tr}\, \widehat{H}_N^2 - \mathbb{E}\, \mathrm{tr}\, \widehat{H}_N^2 = \frac{1}{M_N} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1). \tag{15}$$

In the fixed-q regime, their coefficient is

$$A(f) = \int_{\mathbb{R}} \frac{x f'(x)}{2}\, \rho_\infty(\mathrm{d}x),$$

and their proof amounts to showing that the centered statistic for $f - A(f)x^2$ has variance $o(M_N^{-1})$. Since $\sqrt{M_N} Q_N \Rightarrow \mathcal{N}(0,2)$, the radial term gives their limit.

There are two obstructions to substituting (6) directly into that theorem. First, $f_{N,\beta}$ depends on N through $\tau_N \asymp \sqrt{N}$. Second, neither the fixed-function polynomial approximation nor the bounded-derivative estimate in [1] is uniform for this sequence. In particular, the theorem as stated in [1] does *not* imply a CLT for $Z_N(\beta)$.

The modification is to work with the relative partition function Y_N and to replace the limiting coefficient $A(f)$ by the exact finite-N coefficient

$$A_N(\beta) := \frac{\beta}{2}\, \partial_\beta \log F_N(\beta). \tag{16}$$

Indeed, the next lemma proves the exact FTW decomposition

$$Y_N(\beta) - 1 = A_N(\beta) Q_N + \mathcal{R}_N(\beta), \qquad \mathcal{R}_N = \sum_{r \geq 3} P_r Y_N. \tag{17}$$

The new estimate, replacing the fixed-test-function variance computation of FTW, is

$$\mathbb{E}|\mathcal{R}_N(\beta)|^2 = o(N^2/M_N) = o(N^{2-q}). \tag{18}$$

Sections 4.1–4.3 prove precisely (18).

4 Exact finite-N FTW projection

Lemma 4.1 (First and second chaoses)**.** *For every real β,*

$$P_1 Y_N(\beta) = 0$$

and

$$P_2 Y_N(\beta) = \frac{\beta\, \partial_\beta \log F_N(\beta)}{2M_N} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1). \tag{19}$$

Equivalently, since Clifford trace orthogonality gives

$$\operatorname{tr}(\psi_I \psi_J) = \mathbf{1}_{\{I=J\}}, \qquad \operatorname{tr} H_N^2 = \sigma_N^2 \sum_I g_I^2.$$

Thus the second chaos is the L^2 projection of $Y_N - 1$ onto the centered radial statistic $\operatorname{tr} H_N^2 - \mathbb{E} \operatorname{tr} H_N^2$.

Proof. For $s = (s_1, \dots, s_N) \in \{\pm 1\}^N$, define $\chi_s(I) = \prod_{x \in I} s_x$. We first prove in detail that

$$Z_N(\beta; g) = Z_N\big(\beta; (\chi_s(I) g_I)_I\big). \tag{20}$$

To make the Clifford symmetry explicit, let

$$\Gamma \quad \text{be a unitary phase multiple of} \quad \psi_1 \cdots \psi_N.$$

Since N is even, moving ψ_j through the other $N-1$ Majorana generators produces the sign $(-1)^{N-1} = -1$. Hence

$$\Gamma \psi_j \Gamma^* = -\psi_j \qquad (j \in [N]).$$

For each $x \in [N]$, put $U_x = \Gamma \psi_x$. This is unitary, and conjugation by U_x is the composition of conjugation by ψ_x and conjugation by Γ. The Clifford relations give

$$\psi_x \psi_j \psi_x = \begin{cases} \psi_x, & j = x, \\ -\psi_j, & j \neq x. \end{cases}$$

Combining the last two displays, we obtain

$$U_x \psi_j U_x^* = \begin{cases} -\psi_x, & j = x, \\ \psi_j, & j \neq x. \end{cases}$$

Thus conjugation by U_x changes the sign of precisely the generator ψ_x.

Let $S_s = \{x \in [N] : s_x = -1\}$ and let U_s be the product of the unitaries U_x over $x \in S_s$, in any fixed order. Successively applying the preceding conjugation rule yields

$$U_s \psi_j U_s^* = s_j \psi_j \qquad (j \in [N]).$$

If $I = \{i_1 < \cdots < i_q\}$, then conjugation is multiplicative, so the definition $\psi_I = \mathrm{i}^{q/2}\psi_{i_1}\cdots\psi_{i_q}$ gives

$$\begin{aligned} U_s\psi_I U_s^* &= \mathrm{i}^{q/2}(U_s\psi_{i_1}U_s^*)\cdots(U_s\psi_{i_q}U_s^*) \\ &= \left(\prod_{x\in I} s_x\right)\psi_I = \chi_s(I)\psi_I. \end{aligned}$$

Writing $g^{(s)} = (\chi_s(I)g_I)_I$, we therefore have

$$\begin{aligned} H_N(g^{(s)}) &= \sigma_N \sum_{I\in\mathcal{I}_N} \chi_s(I) g_I \psi_I \\ &= \sigma_N \sum_{I\in\mathcal{I}_N} g_I U_s\psi_I U_s^* = U_s H_N(g) U_s^*. \end{aligned}$$

The power-series definition of the exponential then implies

$$e^{-\beta H_N(g^{(s)})} = U_s e^{-\beta H_N(g)} U_s^*.$$

Finally, the normalized trace is invariant under unitary conjugation. Consequently

$$Z_N(\beta; g^{(s)}) = \operatorname{tr}\Big(U_s e^{-\beta H_N(g)} U_s^*\Big) = \operatorname{tr} e^{-\beta H_N(g)} = Z_N(\beta; g),$$

which proves (20).

More precisely,

$$H_\alpha((\chi_s(I)g_I)_I) = \left(\prod_{x=1}^N s_x^{\sum_{I\ni x}\alpha_I}\right) H_\alpha(g).$$

Averaging the Hermite coefficient against all s shows that it vanishes unless $\sum_{I\ni x}\alpha_I$ is even for every x. This parity rule kills the first chaos. In the second chaos it also kills every off-diagonal monomial $g_I g_J$, since the parity condition for such a monomial forces $I = J$. Permutation symmetry shows that the remaining diagonal coefficients agree. Consequently

$$P_2 Z_N(\beta) = b_N(\beta)\sum_I (g_I^2 - 1)$$

for a deterministic scalar b_N.

We now justify the coefficient calculation. For each fixed I, the probabilists' second Hermite polynomial is $g_I^2 - 1$, with

$$\mathbb{E}(g_I^2 - 1)^2 = 2.$$

Since P_2 is an orthogonal projection and the variables g_I are independent, taking the inner product of the preceding formula for $P_2 Z_N$ with $g_I^2 - 1$ gives

$$2b_N(\beta) = \mathbb{E}\big[Z_N(\beta)(g_I^2 - 1)\big].$$

Applying Gaussian integration by parts twice, conditionally on all coordinates other than g_I, yields

$$\mathbb{E}\big[Z_N(\beta)(g_I^2 - 1)\big] = \mathbb{E}\,\partial_{g_I}^2 Z_N(\beta).$$

Summing over the M_N choices of I therefore gives the first equality below. A further conditional Gaussian integration by parts gives the second:

$$\begin{aligned} 2M_N b_N(\beta) &= \sum_I \mathbb{E}\,\partial_{g_I}^2 Z_N(\beta) = \sum_I \mathbb{E}\big[g_I \partial_{g_I} Z_N(\beta)\big] \\ &= \beta\,\partial_\beta \mathbb{E} Z_N(\beta) = \beta F_N'(\beta). \end{aligned}$$

For completeness, the third equality follows pointwise from the exact scaling identity

$$Z_N(\beta; tg) = \operatorname{tr} e^{-\beta H_N(tg)} = \operatorname{tr} e^{-t\beta H_N(g)} = Z_N(t\beta; g),$$

where we used $H_N(tg) = tH_N(g)$. Differentiating at $t = 1$ gives

$$\sum_I g_I \partial_{g_I} Z_N(\beta; g) = \beta\, \partial_\beta Z_N(\beta; g).$$

Finally,

$$\partial_\beta Z_N(\beta; g) = -\operatorname{tr}\Big(H_N(g) e^{-\beta H_N(g)}\Big), \qquad |\partial_\beta Z_N(\beta; g)| \le \|H_N(g)\|\, e^{|\beta| \|H_N(g)\|}.$$

Together with

$$\|H_N(g)\| \le \sigma_N \sum_I |g_I|,$$

the right-hand side has every exponential moment under the finite dimensional Gaussian law. It gives a locally uniform dominating function in β. The same type of bound justifies the preceding Gaussian integrations by parts. Thus differentiation passes through the expectation and $\partial_\beta \mathbb{E} Z_N(\beta) = F_N'(\beta)$. Dividing by the deterministic quantity $F_N(\beta)$ proves (19). □

4.1 Analytic control of the high-order remainder

The next lemma is the step which upgrades the polynomially weak estimate (8) to the precision required for the CLT.

Lemma 4.2 (Analytic suppression of high chaoses)**.** *Let $w \in \mathbb{C}$ satisfy $|w| < R$, and let $G(z)$ be analytic on a neighborhood of $|z| \le R$ with values in a Hilbert orthogonal sum $\mathcal{H} = \bigoplus_{r\ge 1} \mathcal{H}_r$. Write $G_r(z)$ for its projection onto $\mathcal{H}_r$. Suppose that G_r has a zero of order at least r at the origin. Then, for every integer $m \ge 0$,*

$$\sum_{r>m} \|G_r(w)\|_{\mathcal{H}}^2 \le \frac{(|w|/R)^{2(m+1)}}{1 - (|w|/R)^2} \frac{1}{2\pi} \int_0^{2\pi} \left\|G(Re^{i\theta})\right\|_{\mathcal{H}}^2 d\theta. \tag{21}$$

Proof. Write

$$G_r(z) = \sum_{k\ge r} z^k v_{r,k}, \qquad v_{r,k} \in \mathcal{H}_r.$$

Parseval's identity in the angular variable gives

$$A_r(R) := \frac{1}{2\pi} \int_0^{2\pi} \left\|G_r(Re^{i\theta})\right\|_{\mathcal{H}}^2 d\theta = \sum_{k\ge r} R^{2k} \|v_{r,k}\|_{\mathcal{H}}^2 .$$

With $t = |w|/R < 1$, Cauchy–Schwarz yields

$$\|G_r(w)\|_{\mathcal{H}} \le \left(\sum_{k\ge r} t^{2k}\right)^{1/2} \left(\sum_{k\ge r} R^{2k} \|v_{r,k}\|_{\mathcal{H}}^2\right)^{1/2}.$$

Consequently,

$$\|G_r(w)\|_{\mathcal{H}}^2 \le \left(\sum_{k\ge r} t^{2k}\right) A_r(R) = \frac{t^{2r}}{1-t^2} A_r(R).$$

Sum over $r > m$, use $t^{2r} \le t^{2(m+1)}$, and then use orthogonality of the spaces $\mathcal{H}_r$ under the angular integral. □

Lemma 4.3 (Vanishing order)**.** *For every $r \geq 1$, the $L^2(\gamma_N)$-valued analytic function $z \mapsto P_r Y_N(z)$ has a zero of order at least r at $z = 0$.*

Proof. The Taylor coefficient of order k of $Z_N(z; g)$ is

$$\frac{(-1)^k}{k!} \operatorname{tr} H_N(g)^k,$$

which is a polynomial of total degree k in the Gaussian vector g. It has no Gaussian-chaos component of order larger than k. Therefore $P_r Z_N(z)$ is divisible by z^r. Since F_N is deterministic, $F_N(0) = 1$, and F_N has no zero on the disk under consideration, division by $F_N(z)$ preserves this vanishing order. □

Apply Lemma 4.2 to $G(z) = Y_N(z) - 1$ and $G_r(z) = P_r Y_N(z)$. Combining Proposition 2.1 with Lemmas 4.2 and 4.3, for every $0 < |\beta| < R < R_q$ and every $m \geq 0$ we obtain

$$\sum_{r>m} \|P_r Y_N(\beta)\|_2^2 \leq \frac{C_q}{1 - (|\beta|/R)^2} N^{-a} \left(\frac{|\beta|}{R}\right)^{2(m+1)}. \tag{22}$$

4.2 Uniform control of chaoses up to logarithmic order

Lemma 4.4 (Real-temperature derivative bound)**.** *For real β, every multi-index α of total order r satisfies*

$$\left|\partial_g^\alpha Z_N(\beta; g)\right| \leq (|\beta|\sigma_N)^r Z_N(\beta; g). \tag{23}$$

Proof. Label the r differentiations by $I_1, \ldots, I_r$. The iterated Duhamel formula is a sum over $r!$ permutations of integrals over the simplex $\Delta_r = \{t_j \geq 0 : \sum_{j=0}^r t_j = 1\}$, whose volume is $1/r!$. Every integrand, apart from the factor $(|\beta|\sigma_N)^r$, has the form

$$\operatorname{tr}\left(e^{-t_0\beta H_N} \psi_{I_{\pi(1)}} e^{-t_1\beta H_N} \cdots \psi_{I_{\pi(r)}} e^{-t_r\beta H_N}\right).$$

Here and below the Schatten norms are defined using the normalized trace:

$$\|A\|_p = (\operatorname{tr}|A|^p)^{1/p}, \qquad \|A\|_\infty = \|A\|.$$

For $t_j > 0$,

$$\left\|e^{-t_j\beta H_N}\right\|_{1/t_j} = \left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j},$$

with the usual $p = \infty$ interpretation when $t_j = 0$. Noncommutative Hölder, with Schatten exponents $1/t_j$, and $\|\psi_I\|_\infty = 1$, therefore bounds the absolute value of the integrand by

$$\prod_{j=0}^r \left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j} = Z_N(\beta; g).$$

The permutation count cancels the simplex volume, proving (23). The same labeled formula covers repeated coordinates. □

Call a multi-index α *even* if

$$\sum_{I \ni x} \alpha_I \equiv 0 \pmod 2 \qquad \text{for every } x \in [N].$$

The sign-gauge symmetry used in Lemma 4.1 shows that only even multi-indices have nonzero Hermite coefficients.

Lemma 4.5 (Logarithmic-order finite-chaos estimate)**.** *For every fixed real β there is $D = D(q,\beta) < \infty$ such that, for all $r \geq 1$ and all N,*

$$\|P_r Y_N(\beta)\|_2^2 \leq \exp\big(Dr\log(r+1)\big)N^{-ar}. \tag{24}$$

Consequently, for every fixed $A < \infty$,

$$\sum_{r=3}^{\lceil A\log N\rceil} \|P_r Y_N(\beta)\|_2^2 = o(N^{-2a}). \tag{25}$$

Proof. The Hermite coefficient formula, Lemma 4.4, and $F_N(\beta) > 0$ give

$$\begin{aligned}\|P_r Y_N(\beta)\|_2^2 &= \frac{1}{F_N(\beta)^2}\sum_{|\alpha|=r}\frac{1}{\alpha!}\left|\mathbb{E}\partial_g^\alpha Z_N(\beta)\right|^2 \\ &\leq (|\beta|\sigma_N)^{2r}\sum_{\substack{|\alpha|=r\\ \alpha \text{ even}}}\frac{1}{\alpha!}.\end{aligned} \tag{26}$$

An even ordered r-tuple of q-sets has total incidence qr, and every vertex in its union has incidence at least two. Its union therefore has at most $qr/2$ vertices. Since a multi-index α corresponds to $r!/\alpha!$ ordered tuples, we may first choose a v-set containing the union and then choose the ordered tuple. This can count a tuple more than once, which is harmless for the upper bound. Thus

$$\begin{aligned}\sum_{\substack{|\alpha|=r\\ \alpha \text{ even}}}\frac{1}{\alpha!} &\leq \frac{1}{r!}\sum_{v\leq qr/2}\binom{N}{v}\binom{v}{q}^r \\ &\leq \left(\frac{qr}{2}+1\right)\left(\frac{qr}{2}\right)^{qr}N^{qr/2}.\end{aligned}$$

Using $\sigma_N^{2r} = ((q-1)!)^r N^{-(q-1)r}$ in (26) proves (24) after increasing D.

Let $m_N = \lceil A\log N\rceil$. Uniformly for $3 \leq r \leq m_N$, $D\log(r+1) = o(\log N)$. Hence, for all sufficiently large N,

$$\exp\big(Dr\log(r+1)\big)N^{-ar} \leq N^{-3ar/4}.$$

Summing from $r = 3$ to m_N gives $O(N^{-9a/4}) = o(N^{-2a})$. □

4.3 Completion of the modified FTW remainder estimate

Proposition 4.6 (Higher-chaos closure)**.** *For every fixed real β satisfying $0 < |\beta| < R_q$,*

$$\sum_{r\geq 3}\|P_r Y_N(\beta)\|_2^2 = o(N^{2-q}) = o(N^{-2a}). \tag{27}$$

Proof. Choose R with $|\beta| < R < R_q$ and put $t = |\beta|/R < 1$. Choose a constant A such that

$$2A|\log t| > a,$$

and set $m_N = \lceil A\log N\rceil$. Lemma 4.5 controls $3 \leq r \leq m_N$. For the remaining chaoses, (22) gives

$$\sum_{r>m_N}\|P_r Y_N(\beta)\|_2^2 \leq \frac{C_q}{1-t^2}N^{-a}t^{2(m_N+1)} = o(N^{-2a}).$$

Adding the two estimates proves the claim. □

Proof. Let

$$S_N = \frac{1}{\sqrt{2M_N}} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1).$$

The ordinary i.i.d. central limit theorem gives $S_N \Rightarrow \xi \sim \mathcal{N}(0,1)$. By Lemma 4.1,

$$Y_N(\beta) - 1 = c_N(\beta) S_N + R_N(\beta), \qquad c_N(\beta) = \frac{\beta\, \partial_\beta \log F_N(\beta)}{\sqrt{2M_N}}, \tag{28}$$

where $R_N = \sum_{r \geq 3} P_r Y_N$ is orthogonal to S_N. Since $M_N \sim N^q/q!$, (11) gives

$$N^{q/2-1} c_N(\beta) \longrightarrow \sqrt{\frac{q!}{2}}\, \beta \ell_q'(\beta).$$

Proposition 4.6 gives $N^{q/2-1} R_N \to 0$ in L^2. Slutsky's theorem proves (1). Orthogonality in (28) also gives

$$\mathrm{Var}(Y_N) = c_N(\beta)^2 + \|R_N(\beta)\|_2^2,$$

which proves (2). Finally, $\mathrm{Var}(Y_N) = \mathrm{Var}(Z_N)/(\mathbb{E} Z_N)^2$, and the assumed nonvanishing of the limiting coefficient allows one more application of Slutsky, proving (3). □

Remark 4.7. *Since $\ell_q'(\beta) = \beta/q + O(\beta^3)$ at the origin, the nondegeneracy condition in Theorem 1.1 holds for every sufficiently small real $\beta \neq 0$.*

Remark 4.8 (What replaces the non-uniform step in FTW)**.** *The complex second-moment estimate alone is only $O(N^{1-q/2})$ and is too weak for the CLT scale N^{2-q}. The gain comes from using it only for chaos orders above $A \log N$. Analyticity supplies the factor $(|\beta|/R)^{2A \log N}$ there, while the sign-gauge counting estimate is strong enough uniformly up to order $A \log N$. This is exactly the extra uniformity that the N-dependent test function $f_{N,\beta}$ requires beyond the fixed-test-function theorem of [1]. No CLT for $\log Z_N$ and no fluctuation-level assumption about the free energy is used.*

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